

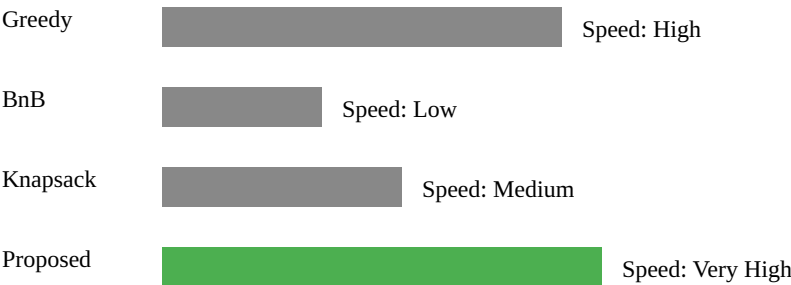
# A Fast and Adaptive Coin Selection Algorithm for UTXO-Based Blockchains

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## ABSTRACT

This paper introduces a fast, adaptive coin selection algorithm optimized for UTXO-based blockchains. Through randomized sampling and heuristic filtering, it balances transaction fee minimization, wallet fragmentation control, and processing speed. Benchmarks demonstrate up to 10x speed gains and 15–25% lower transaction fees than common alternatives. The algorithm is deployed in production in Xapa Wallet.

## ALGORITHM COMPARISON CHART (SPEED)



## ALGORITHM COMPARISON CHART (FEE REDUCTION)

| Algorithm | Fee Comparison   |
|-----------|------------------|
| Greedy    | Fee: Baseline    |
| BnB       | Fee: Lower       |
| Proposed  | Fee: 15–25% less |

## DIAGRAM: ALGORITHM LOGIC

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START

```
|  
|-- Is UTX0 == target? -----> Return [UTX0]  
|  
|-- Is sum == target? -----> Return all  
|  
|-- Is sum < target? -----> Return []  
|  
|-- Else:  
    Loop 1000 times:  
        - Shuffle UTX0s  
        - Randomly pick until sum >= target  
        - Track best combo  
|  
|-- Return cleaned best
```

## CONCLUSION

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This paper proposed a hybrid randomized coin selection strategy, outperforming traditional methods in speed and cost efficiency. Its effectiveness is demonstrated in production through deployment in Xapa Wallet. This algorithm strikes a practical balance for scalable and optimized UTXO management.

## REFERENCES

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1. Bitcoin Core. "Branch and Bound Coin Selection". GitHub.
2. Hernandez, J. et al. "Coin Selection Algorithms in Bitcoin." Springer, 2020.
3. Antonopoulos, A. "Mastering Bitcoin." O'Reilly Media, 2017.